Rongqian Will Chen

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Education

George Washington University (GW) — **Washington, D.C., US**
PhD in Electrical Engineering — *Sept. 2024 – Now*

University of Pennsylvania (UPenn) — **Philadelphia, US**
MSE in Electrical Engineering, GPA: 3.81/4.0 — *Sept. 2021 – May 2023*

Southwest Jiaotong University (SWJTU) — **Chengdu, China**
BEng in Automation, GPA: 3.45/4.0 — *Sept. 2017 – Jun. 2021*

Skills

Mathematical Tools Mathematica, Matlab
Design and Simulation SolidWorks, Ansys, Unity, PSIM
Hardware Development Altium Designer, Keil, Proteus, CCStudio
Embedded Systems Linux, Raspberry Pi, Arduino, STM32, FPGA, DSP
Programming Languages Python, C, C++, C#, Verilog, HTML

Experience

Perception Graph Communication Model for MR Systems — **Washington, D.C., US**
Research Leader — *Aug. 2024 – Present*

- Developed a Perception Graph model that mimics human interpretation in mixed reality, converting MR visual inputs into a compact semantic graph enriched with multi-functional embeddings (probability, contextual weight, spatial relation, text, and graphical features).
- Leveraged pre-trained vision–language models to extract and fuse multimodal features, then built a distortion-distance metric to quantify perception attacks (fake-object insertion, overload, removal) in a latent perception space.
- Integrated Gregory's theory of perceptual information loss and Tulving's contextual influence, enabling context-aware weighting of graph elements for more accurate distortion scoring.
- Employed a hybrid neural-symbolic approach—neural VLM embeddings plus symbolic graph reasoning—to achieve higher interpretability and robustness compared to pixel-level or GNN/GCN-only methods.

AR Guiding System: Assembly King — **Washington, D.C., US**
Lead Developer — *Oct. 2024 – Dec. 2024*

- Developed an AR-guided real-time tutoring system on Meta Quest 3 and PC to assist mechanical part assembly, enabling intuitive coach–trainee interactions.
- Engineered computer-vision and Vision-Language Model pipelines to recognize 3D-printed components, infer assembly steps, and provide context-aware highlighting.
- Implemented real-time instruction overlays and feedback loops, streamlining the assembly workflow and reducing errors.

Macro Economics Bot: MarketDigest — **Online**
Developer — *Apr. 2024 – Sept. 2024*

- Automated scraping and parsing of macroeconomic reports from 14 major institutions (e.g., BlackRock, Goldman Sachs, IMF, ECB, Fed).
- Designed a RAG pipeline: PDF content extraction, LLM-based summarization, and Pinecone vector database for semantic search.
- Built an interactive chatbot interface supporting natural-language queries on market trends, economic indicators, and institutional outlooks.
- Deployed the system using Docker on AWS ECS with S3 for storage; open-sourced code on GitHub.

Pneumatic Legged Hopping Robot (Master's Thesis, Sung Robotic Lab & Kodlab at UPenn) — **Philadelphia, US**
Leader — *Sept. 2022 – July 2024*

- Modeled, fabricated, and characterized a tunable-stiffness pneumatic bellows actuator without external pressure, achieving a 143% stiffness adjustment range via analytical, data-driven, and FEA methods.
- Designed and built a legged hopper robot's mechanical and electrical systems, integrating Raspberry Pi, Pico, Arduino, ESP32 controllers, and a brushless motor controller, plus communication, perception, and data-processing modules.
- Developed a five-bar-linkage ground emulator and its stiffness/damping controller.
- Simulated and validated ground-adaptation energy-saving strategies, demonstrating a 29.3% reduction in energy loss by tuning leg stiffness.

Tunable-Stiffness Coiled-Spring Actuator **Philadelphia, US**
Core Member *Oct. 2021 – Sept. 2022*

- Developed mathematical models for elastic ring deformation and coil stiffness, simulating multilayer spring force–displacement curves.
- Designed and optimized 3D-printed mechanical structures, transmission mechanisms, and electronic circuits; programmed the coil-spring motor controller.
- Applied the actuator to a flexible manipulator, measured workspace with a motion-tracking system, and implemented an end-effector path-tracking controller.

Moving Target Tracking Algorithm of mmWave Radar **Chengdu, China**
Leader *Nov. 2020 – Jul. 2021*

- Deployed and calibrated a 77 GHz FMCW radar data-collection system through road testing.
- Developed and optimized signal-processing algorithms to enhance image resolution and reduce detection errors.
- Designed a robust moving target tracking algorithm using limited radar features.

Pulse Power Converter with Dual-Inductance Active Storage Unit **Chengdu, China**
Core Member *Sept. 2019 – Aug. 2020*

- Designed control circuits using op-amps and improved hysteresis current-control strategies.
- Reduced current spikes and bus voltage ripples through hardware refinements, improving pulse current compensation and lowering power loss.
- Simulated the system in PSIM and validated theory with an experimental prototype.

Acquiring and Stacking Blocks Solution for Robotic Arm **Philadelphia, US**
Core Member *Mar. 2022 – May 2022*

- Calculated optimal stacking trajectories and optimized algorithms to accelerate robotic arm computations.
- Developed pick-up strategies to maximize stacking height and minimize cycle time under variable conditions.

FPGA-Based Acceleration of Deep Convolutional Neural Networks **Philadelphia, US**
Leader *Nov. 2021 – Dec. 2021*

- Transformed convolution operations into parallel matrix multiplications, boosting performance and memory bandwidth.
- Deployed PyTorch models on FPGA, developing custom layers in C and Python.

Embedded AI on Raspberry Pi **Online**
Member *Jul. 2020 – Oct. 2020*

- Built and trained lightweight image-classification models on Raspberry Pi using transfer learning, improving inference accuracy.

Publications

- **Rongqian Chen**, Jun Kwon, Kefan Wu, Wei-Hsi Chen. Tunable Stiffness for Energy-Efficient Vertical Hopping in a Monopedal Robot Across Varying Ground Profiles. *ICRA 2025*.
- **Rongqian Chen**, Jun Kwon, Wei-Hsi Chen, Cynthia Sung. Design and Characterization of a Pneumatic Tunable-Stiffness Bellows Actuator. *RoboSoft 2024*.
- Shivangi Misera, Mason Mitchell, **Rongqian Chen**, Cynthia Sung. Design and Control of a Tunable-Stiffness Coiled-Spring Actuator. *ICRA 2023*.
- **Rongqian Chen**, Yingquan Zou, Anyong Gao, Leshi Chen. A Cluster-Based Weighted Feature Similarity Moving Target Tracking Algorithm for Automotive FMCW Radar. *VTC 2022-Spring*.
- Ping Yang, Xi Chen, **Rongqian Chen**, et al. Stability improvement of Pulse Power Supply with dual-inductance Active Storage Unit Using Hysteresis Current Control. *IEEE JETCAS, 2021*.